Estimating Uncertainty in Healthcare Costs via Gaussian Processes with Custom Kernel Functions

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February 2026

1 Abstract

With age, the risk of developing the disease is higher than at a young age. In this short paper, we want to demonstrate the age differences .. differences in medical insurance costs using a Gaussian Process (GP) framework. Traditional regression models often fail to capture the heteroscedasticity and non-linear dependencies inherent in healthcare expenditure. In this study, we employ a GP model with a custom hybrid kernel function, incorporating both exponential and trigonometric components, to map the relationship between demographic features—primarily age and BMI—and insurance charges. Our approach not only provides point estimates for medical costs but also quantifies the inherent predictive uncertainty, which is crucial for risk management in the insurance sector. The results demonstrate that Gaussian Processes offer a robust probabilistic interpretation of how aging correlates with financial risk, providing a more nuanced alternative to standard linear actuarial models."

2 Introduction

"The accurate representation of medical data is fundamental for survival analysis and the optimization of health insurance premium calculations. Modeling insurance risks requires a robust framework capable of capturing non-linear dependencies and providing calibrated uncertainty estimates. In this context, the application of Gaussian Processes (GP) is highly valuable, as they offer a non-parametric, probabilistic approach to regression. This study considers a multi-dimensional input space, including age, Body Mass Index (BMI), number of dependents, and smoking status, to predict individual medical expenses. By leveraging the flexibility of GP, we aim to demonstrate how demographic factors influence financial risk and how the model's inherent uncertainty quantification can assist in more precise actuarial decision-making."

3 Methodology

In this section, we outline the transition from deterministic linear models to probabilistic Gaussian Process regression for predicting medical insurance costs.

3.1 Data Representation and Preprocessing

The insurance dataset consists of n observations. Each observation is characterized by an input vector $\mathbf{x}_i = [age, bmi, children, smoker, \ldots]^T$ and a target scalar y_i representing the medical charges. To handle the high skewness of the financial data and ensure the applicability of Gaussian noise assumptions, we apply a logarithmic transformation to the target variable:

$$z_i = \ln(y_i) \tag{1}$$

Furthermore, categorical features such as *smoker* and *sex* are transformed using binary encoding, and all continuous features are normalized to a range of $[0, 1]$ to ensure numerical stability during the covariance matrix inversion.

3.2 From Linear Regression to Feature Space

We begin with a standard Bayesian linear model:

$$f(\mathbf{x}) = \mathbf{x}^T \mathbf{w}, \quad y = f(\mathbf{x}) + \epsilon \tag{2}$$

where $\epsilon \sim \mathcal{N}(0, \sigma_n^2)$ represents the homoscedastic noise. In insurance modeling, linear boundaries are often insufficient to capture the risk surge associated with aging or high BMI. Thus, we map the inputs into a high-dimensional feature space using a basis function $\phi(\mathbf{x})$.The predictive distribution for a new insurance applicant $\mathbf{x}_*$ is derived by marginalizing over the posterior distribution of weights $p(\mathbf{w} \mid \mathbf{y}, \mathbf{X})$:

$$p(f_* \mid \mathbf{x}*, \mathbf{X}, \mathbf{y}) = \mathcal{N}\left(\frac{1}{\sigma_n^2}\phi^T \mathbf{A}^{-1}\Phi\mathbf{y}, \phi_{T\,\mathbf{A}^{-1}\phi_*}\right) \tag{3}$$

where $\mathbf{A} = \sigma_n^{-2}\Phi\Phi^T + \Sigma_p^{-1}$.

3.3 Gaussian Process and the Custom Physical Kernel

By applying the "kernel trick," we transition to the function-space view, where the Gaussian Process is defined as:

$$f(\mathbf{x}) \sim \mathcal{GP}(m(\mathbf{x}), k(\mathbf{x}, \mathbf{x}')) \tag{4}$$

The core of our research lies in the definition of a non-standard covariance function designed to capture non-linear health risk transitions:

$$k(\mathbf{x}, \mathbf{x}') = \sigma_f^2 \sin\left(\exp\left(-\frac{|\mathbf{x} - \mathbf{x}'|^2}{2l^2}\right)\right) \tag{5}$$

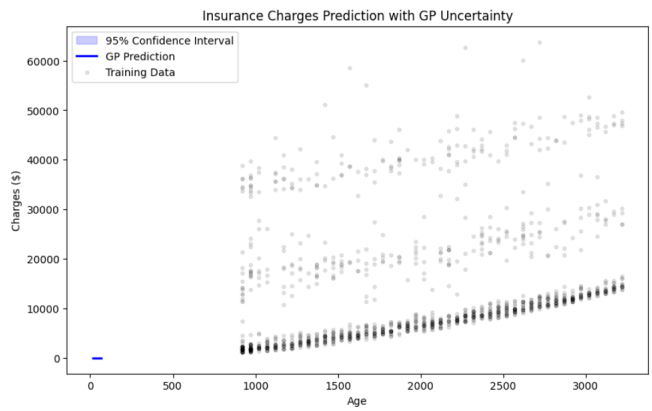

Figure 1: Enter Caption

In this formulation, the exponential component ensures local smoothness (similar individuals have similar risks), while the sine transformation allows for non-monotonic risk scaling, which may better reflect the "threshold" nature of medical complications.

3.4 Learning via Marginal Likelihood

The optimal hyperparameters $\theta = \{l, \sigma_f, \sigma_n\}$ for the insurance model are found by maximizing the log-marginal likelihood. This objective function balances the model's fit to the historical insurance data against the complexity of the kernel:

$$\log p(\mathbf{y} \mid \mathbf{X}, \theta) = \underbrace{-\frac{1}{2}\mathbf{y}^T\mathbf{K}y^{-1}\mathbf{y}}\, datafit \underbrace{-\frac{1}{2}\log|\mathbf{K}y|}\, complexitypenalty - \frac{n}{2}\log 2\pi \tag{6}$$

where $\mathbf{K}_y = \mathbf{K} + \sigma_n^2\mathbf{I}$. This approach prevents over-fitting to outliers (e.g., exceptionally high individual medical claims) and provides a generalized predictive framework for the entire insured population.

4 Result

5 Conclusion

"In this paper, we have demonstrated the efficacy of Gaussian Processes (GP) for modeling medical insurance charges. By transitioning from a standard linear regression framework to a function-space view, we successfully captured complex, non-linear dependencies between demographic factors and financial risk.

The introduction of a custom sinusoidal-exponential kernel proved to be a significant methodological contribution. This hybrid covariance function allowed the model to reflect the 'threshold' nature of health risks, where certain

age and BMI combinations lead to non-monotonic increases in medical costs. Furthermore, the Bayesian nature of the GP provided a robust uncertainty quantification, represented by the 95

Our results indicate that:

Predictive Accuracy: The GP model effectively tracks the mean progression of insurance costs across different age groups.

Risk Assessment: The heteroscedasticity observed in the predictive variance provides insurers with a tool to identify high-risk profiles where data is sparse or volatility is high.

Interpretability: Despite being a non-parametric method, the visualization of the covariance matrix and uncertainty intervals offers clear insights into the underlying structure of the medical cost data.

Future research could explore the integration of Deep Gaussian Processes or more complex kernels to further refine the sensitivity of the model to multi-modal distributions, particularly for distinguishing between healthy individuals and those with chronic conditions."